

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Master of Science in Information Technology (M.Sc.(IT))
Semester-I University Examination November-December 2018
MS 111 Object Oriented Programming in Java

Date: 01.12.18, Saturday

Time: 10.00 a.m. to 01.00 p.m.

Maximum Marks: 70

Instructions:

1. Attempt both sections in separate answer books.
2. Figure to the right indicates marks.
3. Make a suitable assumption when necessary.

SECTION – I**Q1 Do as directed. [11]**

1. Wrapping up of data and functions together into a single unit is known as _____.
a) Polymorphism b) Encapsulation c) Inheritance d) Abstraction
2. _____ is the intermediate representation of Java source code which is produced by the Java compiler by compiling the source code.
a) Object code b) Byte Code c) Executable Code d) None of these
3. Which of the following operators can operate on a boolean variable?
a) +,- b) %,/ c) =, += d) &&, ==
4. Which of these jump statements can skip processing the remainder of the code in its body for a particular iteration?
a) break b) continue c) return d) exit
5. Write output of executing `System.out.println( (15 & 14)+ " - " + (12|16));`
a) 14 - 28 b) 15 - 26 c) 28 - 14 d) 26 - 15
6. Which of these is an incorrect array declaration?
a) `int a[]=new int[5]` b) `int[] a=new int[5]` c) `int a[]=new [5]int` d) `int a[]={1,2,3,4}`
7. Write value of `x` after executing `int x=0,y=0,z=0; x=(++x + y--)*z++;`
a) -1 b) 3 c) 0 d) 1
8. Which of these method of String class can be used to test strings for equality?
a) `isequal()` b) `equals()` c) `isequals` d) `equal()`
9. Write output of executing `System.out.print(Math.ceil(10.16));`
a) 10 b) 9 c) 11 d) 10.16
10. Write output of executing `System.out.println("Hello Students".charAt(7));`
a) S b) t c) u d) Error
11. private access specifier can be used with the main method. State True/False.

Q2 Answer the following questions.

- [A] Explain the default and parameterized constructors with and without the use of this keyword. [06]
- [B] What is inheritance? List and explain the types of inheritance. [06]

OR**Q2 Answer the following questions.**

- [A] Differentiate between Abstract class and Interface. [06]
- [B] Explain the inner class and anonymous class in java with example. [06]

Q3 Answer the following questions. (Any Three)

- [A] List the main concepts of OOP and explain any three concepts in brief. [12]
- [B] Explain the use of super keyword.
- [C] List the access specifiers used in java. Explain their use with suitable examples.

- [D] Show the use of package and import keywords taking suitable example.
[E] What is Boxing and Unboxing? Explain with suitable example.

SECTION – II

Q4 Do as directed. [11]

1. Which of these packages contain classes and interfaces used for input & output operations of a program?
a) java.awt b) java.lang c) java.io d) java.util
2. Which of this keyword must be used to inherit a class?
a) implements b) extends c) this d) super
3. Which of these method of Object class can generate duplicate copy of the object?
a) clone b) copy c) replicate d) duplicate
4. Write output of executing `System.out.println("hello".lastIndexOf('l'));`
a) 0 b) 2 c) 3 d) 4
5. Which of these packages contain all the collection classes?
a) java.util b) java.lang c) java.net d) java.awt
6. Which of these class is a superclass of all other classes?
a) Process b) System c) Object d) None of these
7. Which methods are provided by Collections utility class?
a) sort b) shuffle c) reverse d) All of these
8. Which of these class can generate an array which can increase and decrease in size automatically?
a) MallocList b) ArrayList c) LinkedList d) DynamicList
9. Which of these methods can be used to move to next element in a collection?
a) next() b) move() c) shuffle() d) hasNext()
10. Which of these keywords is used to manually throw an exception?
a) try b) finally c) throw d) catch
11. Method overriding is combination of inheritance and polymorphism. State True/False.

Q5 Answer the following questions.

- [A] What is exception handling? Explain try, catch and finally block with example. [06]
[B] Write a code to create an ArrayList and display it using Iterator interface. [06]

OR

Q5 Answer the following questions.

- [A] Explain any six built-in exceptions with the code to generate the exception and handle the exception. [06]
[B] Write a code to open a text file , read and display its content. [06]

Q6 Answer the following questions. (Any Three)

- [A] Explain life cycle of a thread. [12]
[B] What is the use of io package? Explain the use of BufferedReader Class.
[C] Explain the use of throws clause in Exception handling.
[D] What is event handling in java? List any two event classes and listener interfaces.
[E] What is an Applet? Explain the advantages and disadvantages of using Applet.
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